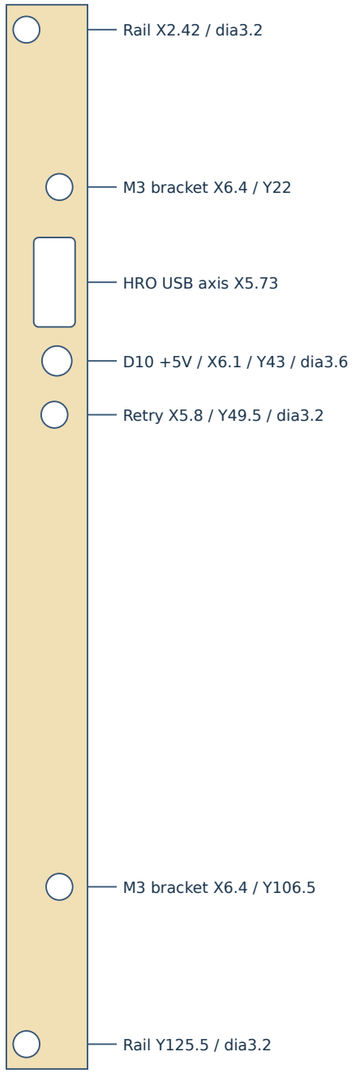
V1 TOP USB / SINGLE POWER LED | TWO BOARDS + TWO PLAIN M3 L BRACKETS

Front panel + perpendicular main PCB. No middle board, spacers or UI headers. mm. ROUTED HRO CARRIER; BARE-PCB ARTWORK, NOT BENCH QUALIFICATION.

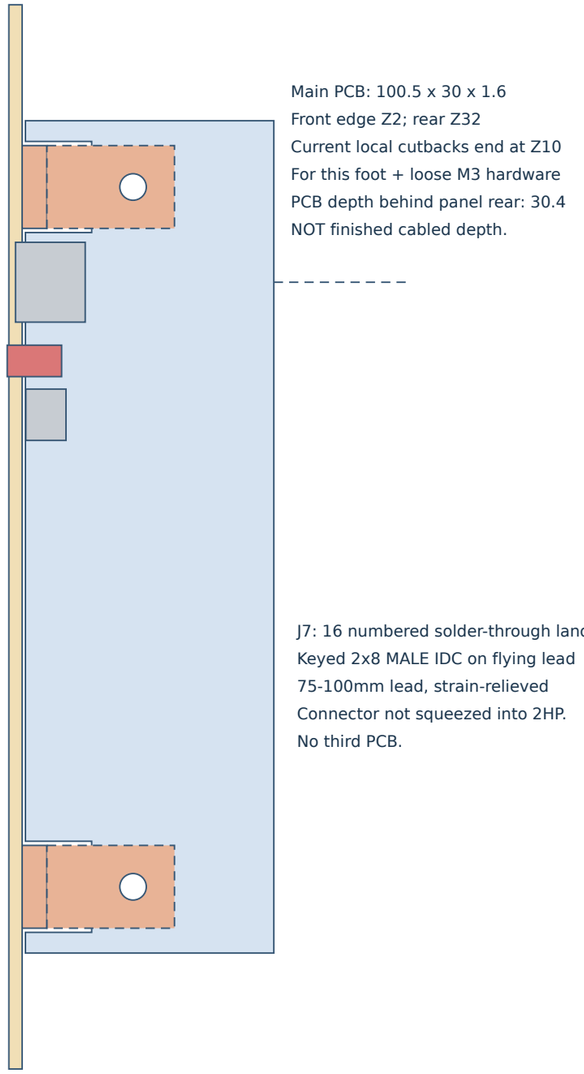
FRONT PANEL



9.8 x 128.5 x 1.6 FR4
Round corner rail holes retained.
6mm rail screw heads overhang .58mm.
Check adjacent panels; no countersink.

0.60mm panel shift: aperture X3.30..8.30; D10 X6.10; SW1 X5.80.
Carrier and bracket holes unchanged; bracket PCB web2.40mm +0.0/-0.1.

SIDE / YZ ELEVATION

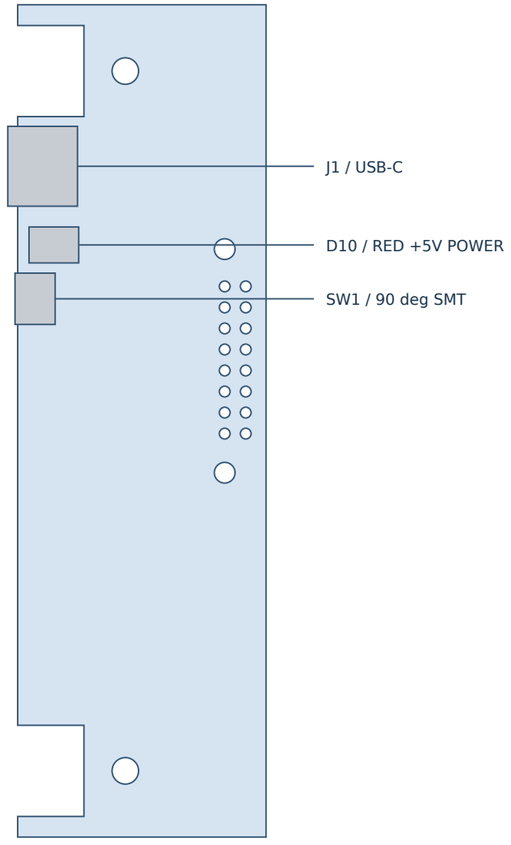


Main PCB: 100.5 x 30 x 1.6
Front edge Z2; rear Z32
Current local cutbacks end at Z10
For this foot + loose M3 hardware
PCB depth behind panel rear: 30.4
NOT finished cabled depth.

J7: 16 numbered solder-through lands
Keyed 2x8 MALE IDC on flying lead
75-100mm lead, strain-relieved
Connector not squeezed into 2HP.
No third PCB.

HRO land imported; mouth Y33.5 / Z0.8; NO 3D model.
PCB BACK X2.5 / component face X4.1.
Web on PCB BACK; foot on front panel REAR (Z1.6).
Screws normal to each face; PCB not fed through L.
Uncut FR4/foot overlap: X2.5..4.1, Z2..4.6.
Y17..27 / 101.5..111.5; current configuration only.
Cutbacks clear THIS L + M3 nut/tool access.
Z10 retained, not a proven minimum.
90-degree face mounts do not inherently need notches.
Notch-free 2HP here needs changed hardware; not selected.
USB and LED/reset openings follow the shown axes.

MAIN PCB / COMPONENT SIDE



D10 ONLY: RED 3mm +5V POWER THT, formed 90 deg.
SW1: E-Switch TL3340AF160QG, SMT.
Three signal pads + two solder anchors.
Front-side fine-tip soldering; 2 locating holes.
3.3mm body height; axis X6.4, recessed Z2.05.
Pins1+3 common; pin2 other contact. No plunger.

12 V power output pin pairs:
1-2: -12V | 3-8: GND | 9-10: +12V
11-12: +5V | 13-16: NOT CONNECTED
POWER OUTPUT, not a rack power input!
Do not connect to another active PSU.

Routed HRO. D10: P5 -> R32 1k -> LED -> GND. No blink/controller. L1 max5.0mm (TMPC1205HP) with5.40mm envelope; L4 max4.0; L2/L3 max4.7. Physical trial/bench validation pending.

HRO TYPE-C-31-M-12 | USB FIT AND DATUM REVIEW

Exact Bread local lands + primary HRO body drawing. NO verified STEP model. This is a dimensioned envelope, not physical qualification.

Panel X across / Y down / Z into case. Carrier F.Cu X4.1; board front Z2.

HRO shell: 8.94 wide x 7.35 deep x 3.26 high (drawing). Nominal shell axis X5.73.

Mating face Z0.8, Y33.5. Native footprint anchor (69.25,103.475), rotation 0 degrees.

Datum anchors: front shell tab 2.60 from face; rear 6.77 drawing / 6.78 local.

Ambiguous drawing vertical 0.50/1.64/4.18/5.79 chain is NOT used to invent a PCB-edge datum.

Local pad row: 7.695 from mouth; local land length 1.45, not claimed manufacturer-dimensioned.

Old GCT land row was 7.355 from mouth: HRO moves row 0.34 rearward; shell slot centres stay.

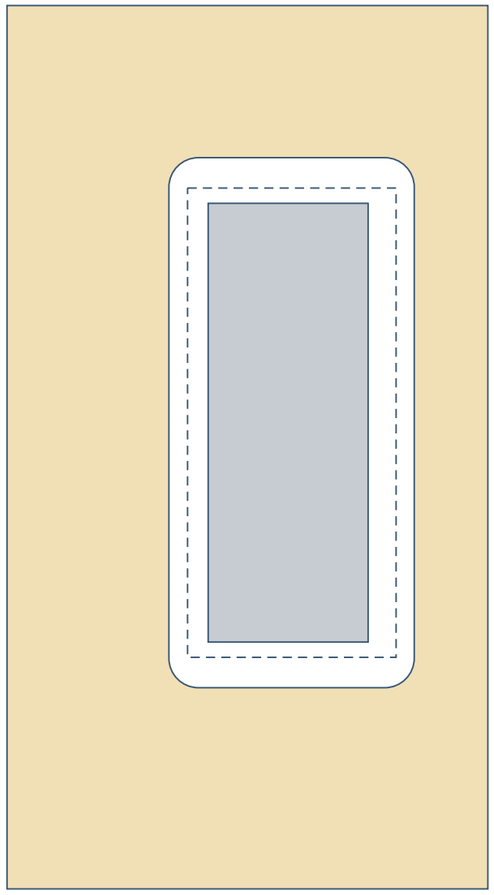
Local bosses: dia0.65 NPTH, nominal molded boss dia0.50; no peg shrinking.

Local shell drills: two 0.6 x1.7 rear and two 0.6 x1.2 front plated slots; no shell paste.

Front slot: nominal drill-to-board-edge 0.80; copper-to-board-edge 0.60.

A1/B12, A4/B9, A9/B4, A12/B1 share four wide PCB lands. S1 shell is GND.

A5=CC1, B5=CC2; all four data and two SBU contacts remain explicit NC.



FRONT VIEW: GRAY NOMINAL / DASHED WORST-CASE SHELL

Aperture: X3.30, Y28.10, W5.00, H10.80, R0.60 mm.

Delta from 024: X -0.93, Y -0.50, width +0.80, height +1.00.

Each machined wall +/-0.10; centre registration +/-0.10 X/Y.

Corner radius 0.50..0.70; profile acceptance THROUGH thickness.

BOTH finished boards 1.60 +/-0.16 mm, including coating.

Carrier back-face datum fixed: thickness error moves shell in X.

Assembly registration +/-0.10 X/Y; seating +0.05 X.

Shell width +0.15, height +0.10; all errors add worst-case.

Squareness <=0.5 deg; +/-0.02 tilt sweep through max1.76 panel.

Minimum modeled corner clearance: 0.233 mm.

In-plane angular registration <=0.5 deg; sweep +/-.040 X / .016 Y.

Assembly registration must be measured; M3 freeplay is not proof.

Panel9.8 x128.5 / carrier100.5 x30; D10/SW1/J7/brackets fixed.

Mouth recess 0.42..1.18; ahead of rear face 0.58..1.02 mm.

Cable insertion/overmould, ABS and actual assembled trial pending.

Bare-PCB aperture fit is NOT product/strength/bench qualification.

Fabrication process requirements: docs/fabrication.md.